

4.Troubleshooting

4-1. Checkpoints by Error Mode

Oscilloscope Setting Values	
Voltage/DIV	1V/div
TIME/DIV	500ms/div

4-1-1. No Power

Symptom	- If you connect the power cables, the LED on the front will turn on - If the units do not work
Major checkpoints	- If the cable is improperly connected or the motherboard does not work, the LED on the front is not working. In this case, check the following. - Check the cable connection. - Check the fuse of the components.
Diagnostics	The Power does not Turn on
Caution	Before working, disconnect the power cord.

4-1-2. No Display

Symptom	- If you connect the power cables, the LED on the front will turn on - If the units do not work or DisplayIncorrect
Major checkpoints	- If the cable is improperly connected or the motherboard does not work, the LED on the front is not working. In this case, checkthe following. - Check the cable connection. - Check the fuse of the components.
Diagnostics	The Power does not Turn on ↓ YES Check if 24V is measured on the main pcb F903 NO Check if 24V is measured on the SMPS pin #1 NO Check the AC Cord output NO Replace the AC cord YES Main PCB F903 change YES Check the SMPS and replace SMPS Check ih "ALIVE3V3" is measured at the sym bol FB510 when main pcb power is turned on NO Replace the IC903 YES Check the OLED display NO Replace the OLED display YES Change the MAIN PCB
Caution	Before working, disconnect the powercord.

4-1-3. No Sound Output

Symptom	• If the sound is not output.
Major checkpoints	• Speaker connector is not connected properly or check for damage. • Make sure your sound processing functions operate properly on the motherboard. • Make sure the speaker isn't bad.
Diagnostics	No output YES Check if +3.3V power is measured at FB118 of U100 on the Main PCB. NO Change Main PCB YES Check if 3.3V power is measured at pin1 of JK602 on the HDMI PCB NO Main to HDMI connection check NO Change HDMI PCB YES Check if signal is measured at Pin No.16 of IC5001 IC5002 on the Main PCB. NO Replace IC5001 IC5002 NO Change Main PCB YES Check if signal is measured at pin No. 1,2,4,5,44,45,47,48 of IC5001 IC5002 on the Main PCB NO Main to HDMI connection check NO Change Main PCB YES Check the Main PCB CN5001 CN5002 speaker output signal NO Change speaker connector NO Change Main PCB YES Change speaker L/R Check SPDIF output Check DSP output Check Speaker output
Caution	• Before working, disconnect the power cord.

■ Location of Parts

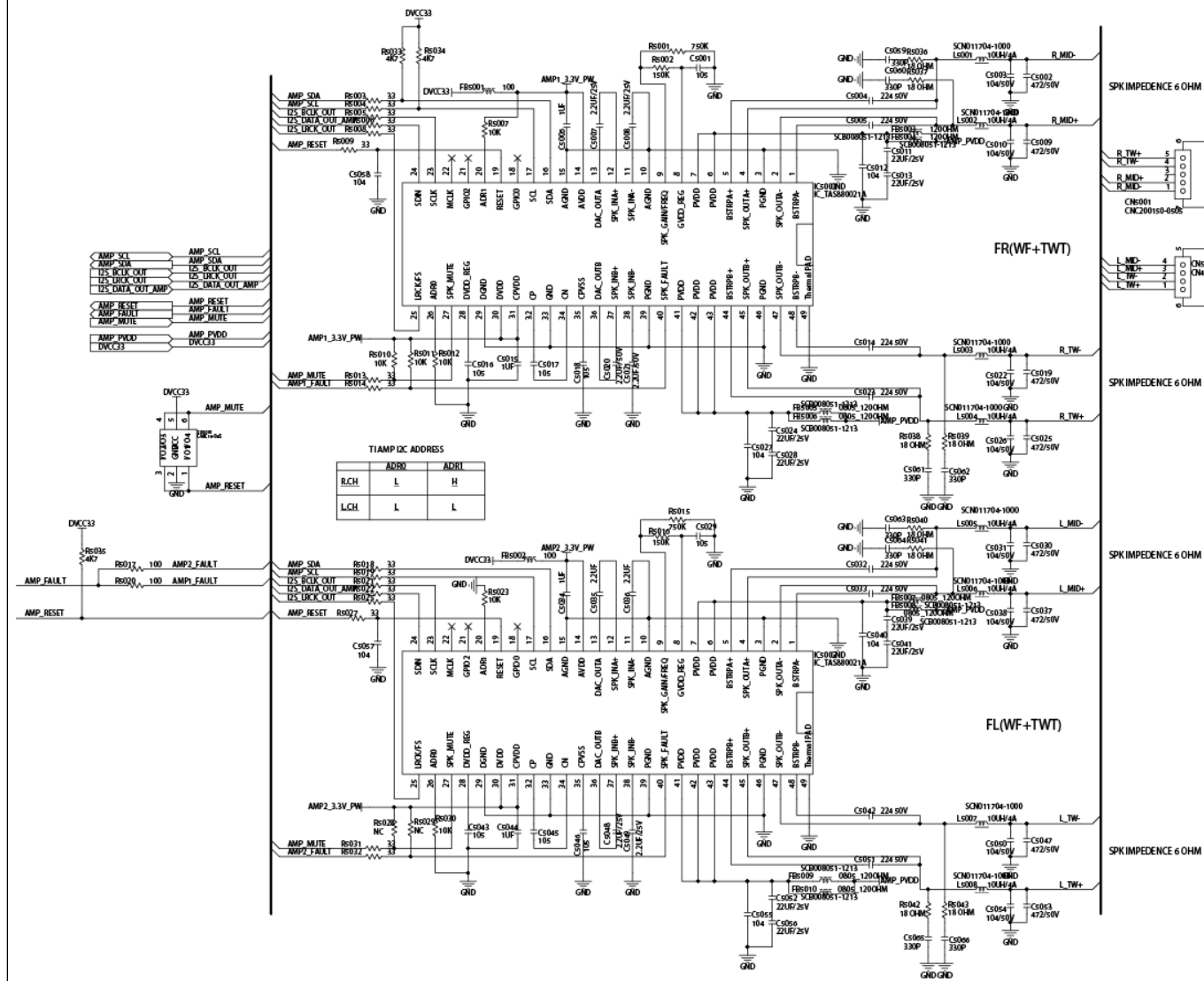


5) CN5002
L_SPEAKER

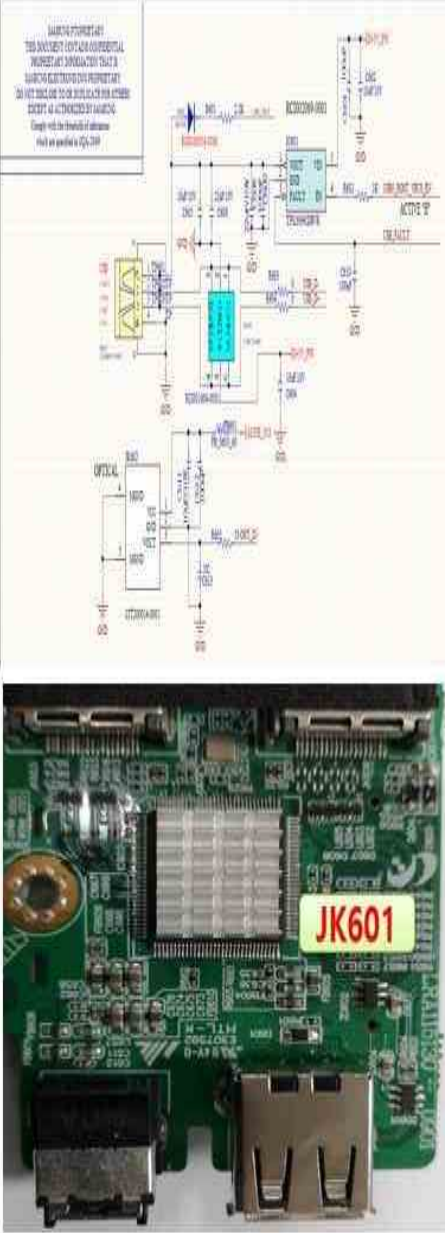
Pin.no	signal
1	L_TW+
2	L_TW-
3	R_MID+
4	R_MID-

6) CN5001
R_SPEAKER

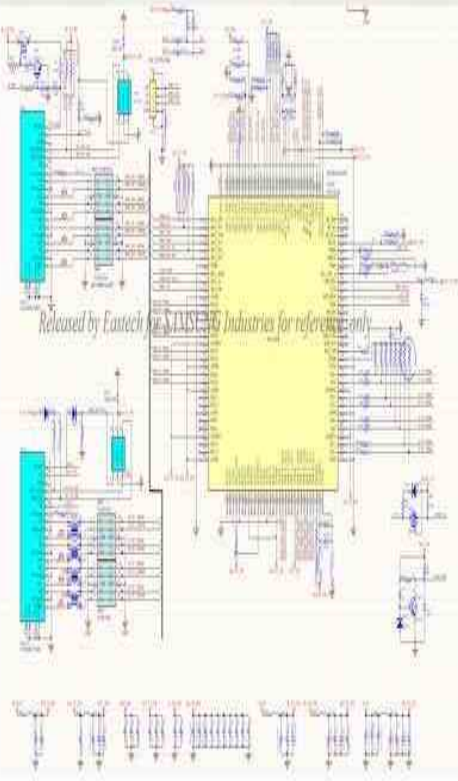
Pin.no	signal
1	R_MID-
2	R_MID+
3	NC
4	R_TW-
5	R_TW+



4-1-4. USB Failure Analysis

Diagnostics	Start YES Check if +5V power is measured at pin 5, 1 of IC601 on the Main PCB, NO Replace IC901pin3 (+5V)Check NO Change MAIN PCB YES Check if +5V power is measured at pin 1 of JK601 on the Main PCB, NO Replace IC601 Check NO Change MAIN PCB YES Check if signal is(USB_D-, USB_D+) measured at pin 2, 3 of JK601 on the Main PCB NO Replace JK601 Check NO Change MAIN PCB YES Check if signal is measured at pin A15, A18 A17 of U100 on the Main PCB, NO Replace U100 Check NO Change MAIN PCB YES END  The image contains two parts: a circuit diagram at the top and a photograph of a printed circuit board (PCB) at the bottom. The circuit diagram shows a USB connector interface with various components including resistors, capacitors, and integrated circuits (ICs). It labels pins for USB_D-, USB_D+, and +5V. The photograph shows a green PCB with a USB connector. A yellow label with the text 'JK601' is placed over a component on the board. The component is a small, rectangular integrated circuit.
Caution	• Before working, disconnect the powercord.

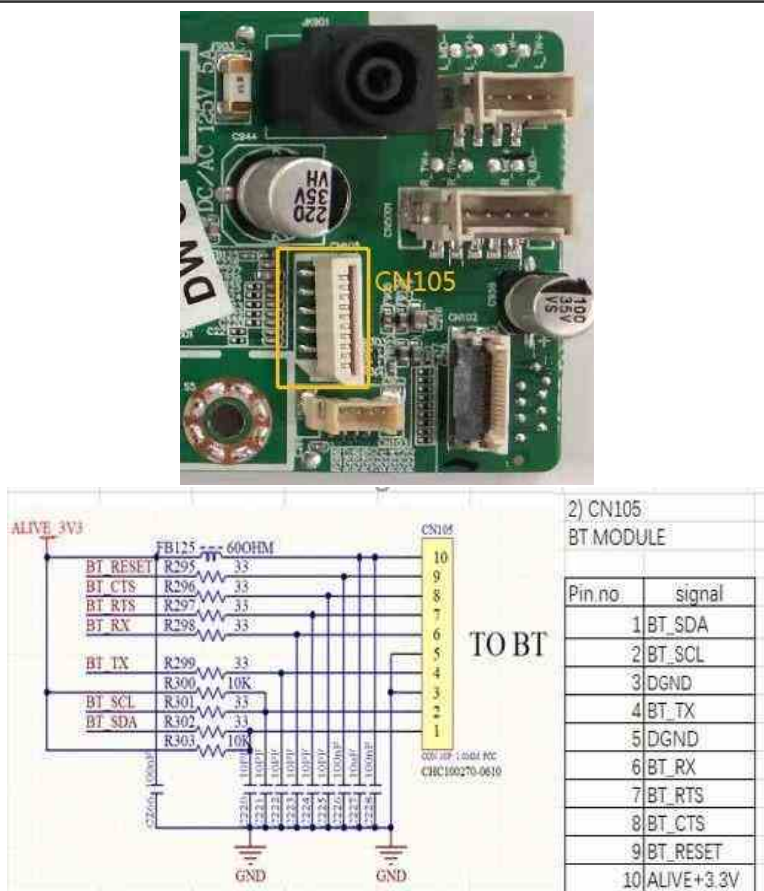
4-1-5. HDMI Failure Analysis

Diagnostics	Start YES Check if H3V_PW power is measured at pin 104, 110, 55 of IC602 on the Main PCB. NO Replace U901 Check NO Change MAIN PCB YES Check if signal HMC0_SDA, HMC0_SCL is measured at pin No. 3, 4 of IC601 on the Main PCB. NO Replace IC601 97, 98 pin Check NO Change MAIN PCB YES Check if signal is measured at pin No. 15, 16 of IC601 on the Main PCB. NO Replace IC601 51, 52 pin Check NO Change MAIN PCB YES Check if signal is measured at pin No. 15, 16 of IC601 on the Main PCB. NO Replace IC601 57, 60 pin Check NO Change MAIN PCB YES Check if signal is H_T2+_HDMI, HS0_F2+_HDMI of IC601 on the Main PCB. NO Replace IC601 Check NO Change MAIN PCB YES END 
Caution	• Before working, disconnect the powercord.

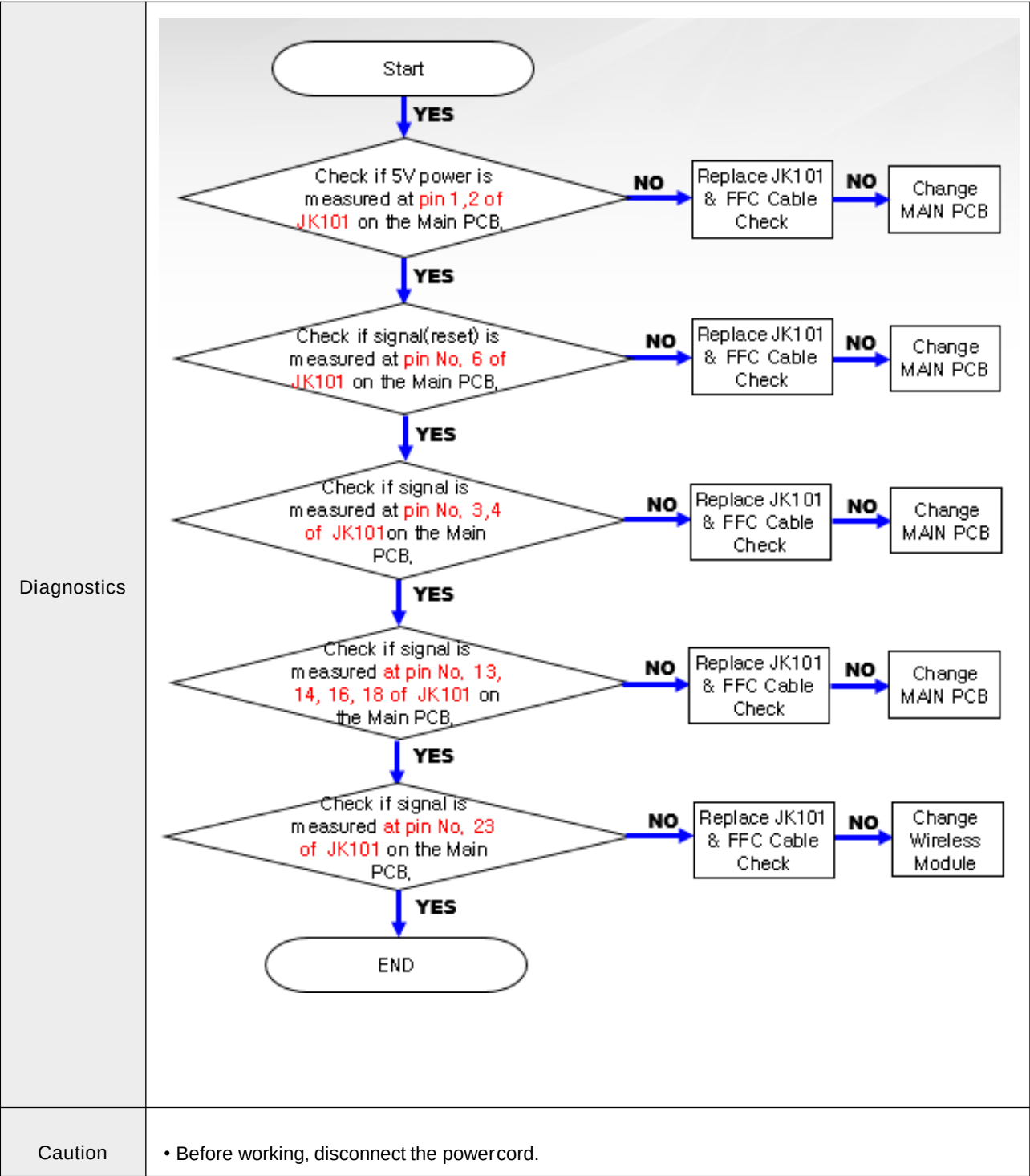
4-1-6. BT Module Failure Analysis

Diagnostics	<pre>graph TD; Start([Start]) -- YES --> D1{Check if 3, 3V power is measured at pin 10 of CN106 on the Main PCB.}; D1 -- NO --> R1[Replace CN106 & wire check]; R1 -- NO --> C1[Change MAIN PCB]; D1 -- YES --> D2{Check if signal(reset) is measured at pin No. 9 of CN106 on the Main PCB.}; D2 -- NO --> R2[Replace CN106 & wire check]; R2 -- NO --> C2[Change MAIN PCB]; D2 -- YES --> D3{Check if signal is measured at pin No. 1, 2 of CN106 on the Main PCB.}; D3 -- NO --> R3[Replace CN106 & wire check]; R3 -- NO --> C3[Change MAIN PCB]; D3 -- YES --> D4{Check if signal is measured at pin No. 4 of CN106 on the Main PCB.}; D4 -- NO --> R4[Replace CN106 & wire check]; R4 -- NO --> C4[Change MAIN PCB]; D4 -- YES --> D5{Check if signal is measured at pin No. 6 of CN106 on the Main PCB.}; D5 -- NO --> R5[Replace CN106 & wire check]; R5 -- NO --> C5[Change BT Module]; D5 -- YES --> End([END]);</pre>
Caution	• Before working, disconnect the powercord.

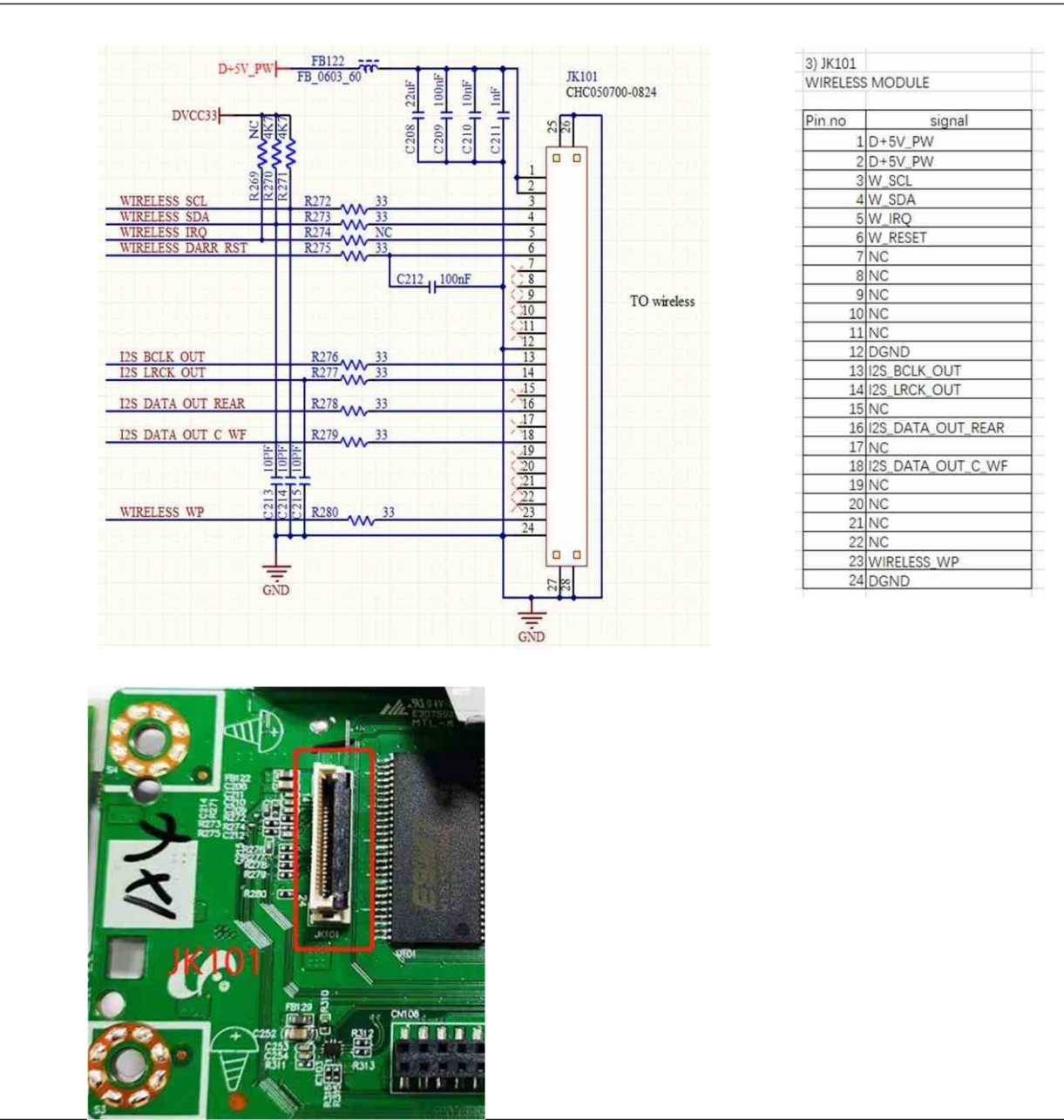
■ Location of Parts



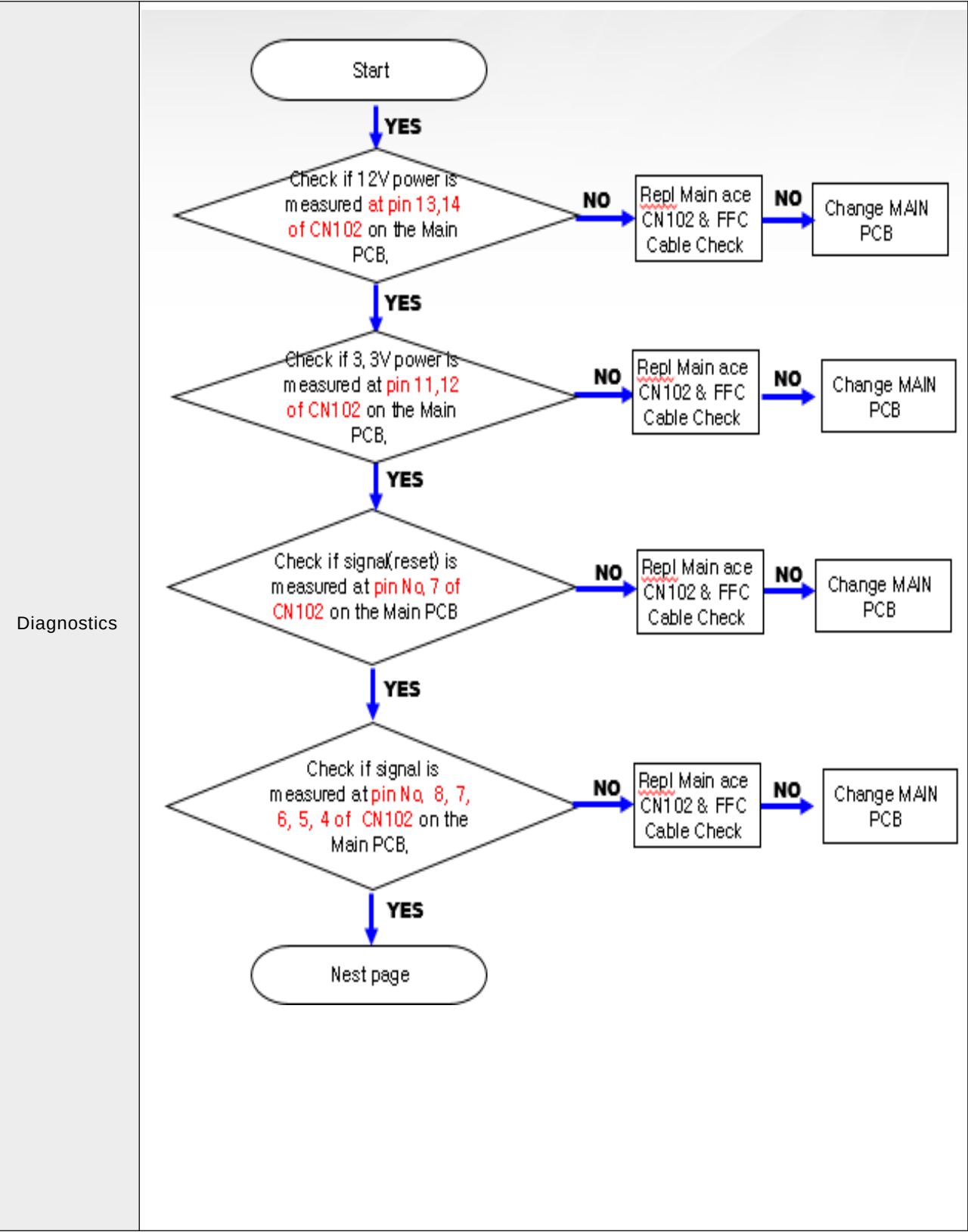
4-1-7. Wireless Module Failure Analysis



■ Location of Parts

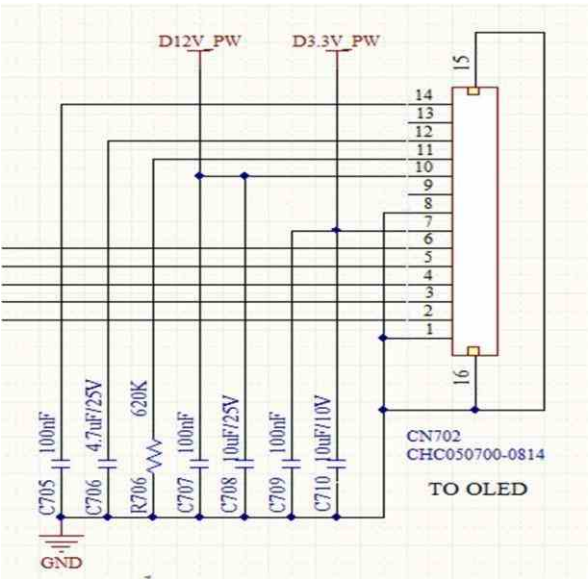


4-1-8. OLED Module Failure Analysis



Diagnostics	<pre> graph TD Start(()) -- YES --> D1{Check if 12V power is measured at pin 10 of CN702 on the Key PCB.} D1 -- NO --> R1[Replace CN 702] R1 -- NO --> C1[Change OLED Module] D1 -- YES --> D2{Check if 3.3V power is measured at pin 7 of CN702 on the Key PCB.} D2 -- NO --> R2[Replace CN 702] R2 -- NO --> C2[Change OLED Module] D2 -- YES --> D3{Check if signal(reset) is measured at pin No. 5 of CN702 on the Key PCB.} D3 -- NO --> R3[Replace CN 702] R3 -- NO --> C3[Change OLED Module] D3 -- YES --> D4{Check if signal is measured at pin No. 2, 3, 4, 6 of CN702 on the Key PCB.} D4 -- NO --> R4[Replace CN 702] R4 -- NO --> C4[Change OLED MODULE] D4 -- YES --> End([END]) </pre>
Caution	• Before working, disconnect the powercord.

■ Location of Parts



5) CN702	
OLED	
Pin.no	signal
1	DGND
2	OLED_SDA
3	OLED_SCL
4	OLED_DC
5	OLED_RESET
6	OLED_CE
7	D+3.3V_PW
8	DGND
9	NC
10	D+12V
11	DGND
12	DGND
13	NC
14	DGND



4-2. Measures to be taken when the Protection Circuit operates

4-2-1. AMP Pre-Inspection relating to Power Protection

If you think there are problems at the AMP PCB, you can check the PCB without disassembling the set by following the test below.

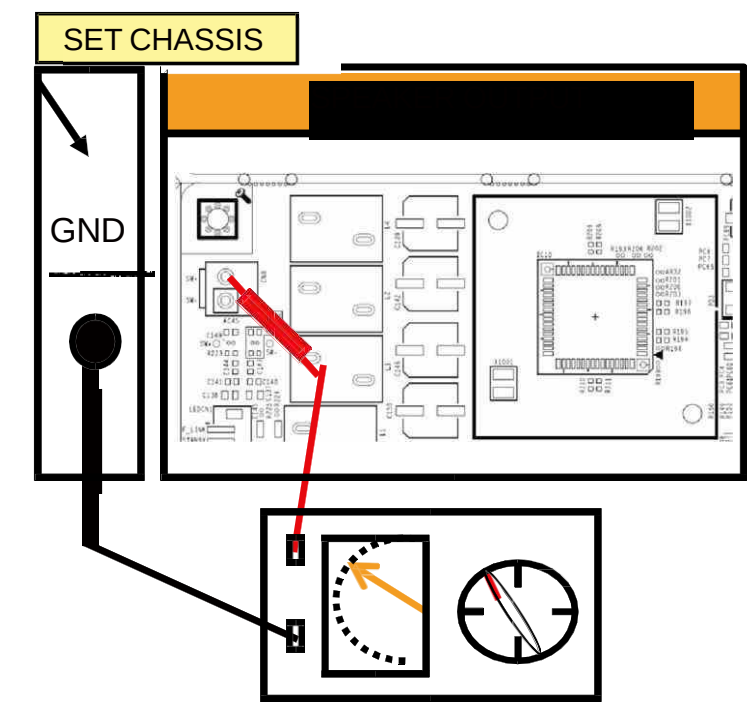


Caution

- Do not connect the power cord during the test!

Resistance using Tester	
F/R CH	137K ohm

- If there is a large difference than the value listed above then the AMP PCB has a problem.



4-3. Initialization & Update

4-3-1. Set Initialization method

■ Sound Bar(Using the remote control)

1. Prepare the remote control.
2. Turn the unit off.
3. Press and hold the “Play/Pause” button on the remote for more than 5 seconds. 
4. If it was successful, “INIT”->”OK” will be displayed for 2 seconds and be turned off.

„ ■ Sound Bar(Without Using the Remote control)

1. Turn the unit on.
2. Press and hold the “+” button and “-” button on Sound Bar simultaneously for 5 seconds. 
3. If it was successful, “INIT”->”OK” will be displayed for 2 seconds and be turned off.
4. The Sound bar will be turned off

4-3-2. Pairing method

„ ■ Sound Bar(Using the remote control)

1. Prepare the remote control.
2. Turn the unit off.
3. Press and hold the “MUTE” button on the remote for more than 5 seconds. 
4. If it was successful, “ID SET” will be displayed for some seconds and be turned off
5. When the soundbar is turned on, it will start finding wireless Rear kit.

„ ■ Sub-woofer

1. You can see the ID-set button the rear side of Sub-woofer.
2. When Sub-woofer is turned on, press the ID-set button for 5 seconds.
3. The STANDBY indicator is turned off and the LINK indicator (Blue LED) blinks quickly.
4. The main unit and the sub-woofer will pair automatically,
- 5 The LINK indicator constant on

„ ■ Rear Kit(Sold separately, SWA-8500S)

1. You can see the ID-set button the rear side of Rear Kit.
2. When Rear Kit is turned on, press the ID-set button for 5 seconds.
3. The STANDBY indicator is turned off and the LINK indicator (Blue LED) blinks quickly.
4. The main unit and the Rear Kit will pair automatically.
- 5 The LINK indicator constan

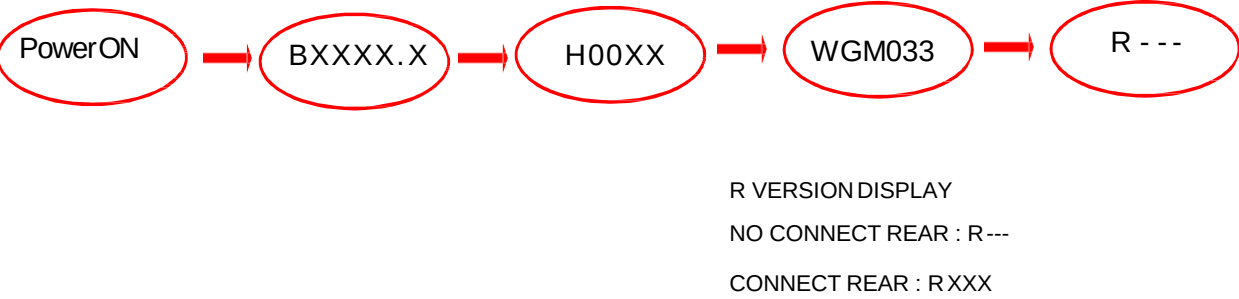
4-3-3. How to check the Firmware version

■ Software check method

- Sound Bar (Remote control use)

- 1. Prepare the remote control.
- 2. Turn the unit off.
- 3. Press the “SOUND CONTROL” button for 5 seconds. 
- 4. OLED will display each versions like below sequence.(Software version → HDMI → Tx → Rx)

Tx : Wireless Tx module in the soundbar
Rx : Wireless Rx module in the rear kit



4-3-4. USB Update procedure

Press the  (Source) button on the top panel or on the remote control, and then select “USB”.

- Step 1 : Power on the device. Change the source to USB, You can see the following OLED display..



- Step2 : Prepare the USB disk with update files then plug-in to the USB port.

You can see the following OLED display.



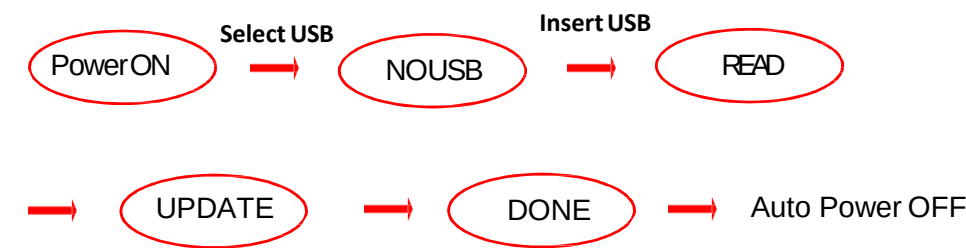
Then power off

- Step3 : Progressingupdate.
In case HDMI and Wirelessupdate, you can see the following OLEDdisplay

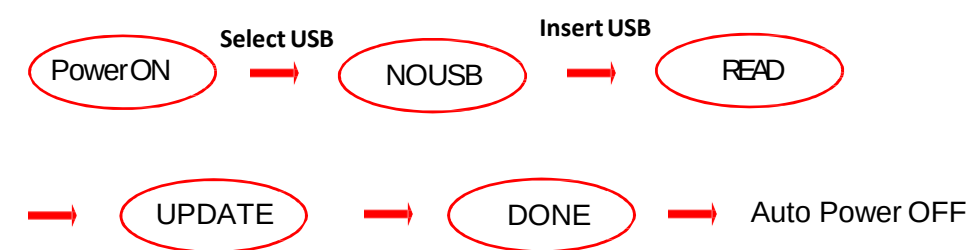
4-3-5. USB Update procedure

Press the  (Source) button on the top panel or on the remote control, and then select "USB".

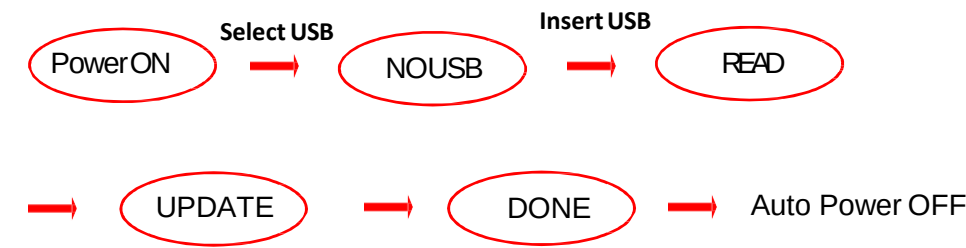
Case1 : DSPUpdate.(HMS-T550WWB-XXXX.X.bin)



Case2 : HDMI Update. (HMS-T550WWH-XXXX.ssp)



Case3 : WirelessUpdate.(HMS-T550WWMR-TXXXX.ssp)



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